

Homework Section 15.5

1. Find the mass and the center of mass of the lamina that occupies the region R and has density $\rho(x, y)$.
 - a) $R = \{(x, y) \mid 0 \leq x \leq 2, -1 \leq y \leq 1\}$; $\rho(x, y) = xy^2$
 - b) R is bounded by the parabola $x = y^2$ and the line $y = x - 2$; $\rho(x, y) = 3$
2. A lamina occupies the part of the disk $x^2 + y^2 \leq 1$ in the first quadrant. Find its center of mass if the density at any point is proportional to its distance from the x -axis (i.e. $\rho(x, y) = ky$, where k is a real constant).
3. Find the moments of inertia I_x , I_y , I_0 for the lamina in exercise 1(b).
4. A lamina with constant density $\rho(x, y) = \rho$ occupies the region under the curve $y = \sin x$ from $x = 0$ to $x = \pi$. Find the moments of inertia I_x and I_y and the radii of gyration $\bar{\bar{x}}$ and $\bar{\bar{y}}$.